```
import pandas as pd

df = pd.read_excel("customer_booking.csv.xlsx")
df.head()
```

	num_passengers	sales_channel	trip_type	purchase_lead	length_of_stay	flight_hour	flight_day	route	booking_origin	w
0	2	Internet	RoundTrip	262	19	7	Sat	AKLDEL	New Zealand	
1	1	Internet	RoundTrip	112	20	3	Sat	AKLDEL	New Zealand	
2	2	Internet	RoundTrip	243	22	17	Wed	AKLDEL	India	
3	1	Internet	RoundTrip	96	31	4	Sat	AKLDEL	New Zealand	
4	2	Internet	RoundTrip	68	22	15	Wed	AKLDEL	India	

Next steps: Generate code with df New interactive sheet

```
df.info()
df.describe()
df.isnull().sum()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 50000 entries, 0 to 49999
Data columns (total 14 columns):
 #   Column                 Non-Null Count  Dtype
---  ------                 --------------  -----
 0   num_passengers         50000 non-null  int64
 1   sales_channel          50000 non-null  object
 2   trip_type              50000 non-null  object
 3   purchase_lead          50000 non-null  int64
 4   length_of_stay         50000 non-null  int64
 5   flight_hour            50000 non-null  int64
 6   flight_day             50000 non-null  object
 7   route                  50000 non-null  object
 8   booking_origin         50000 non-null  object
 9   wants_extra_baggage    50000 non-null  int64
 10  wants_preferred_seat   50000 non-null  int64
 11  wants_in_flight_meals  50000 non-null  int64
 12  flight_duration        50000 non-null  float64
 13  booking_complete       50000 non-null  int64
dtypes: float64(1), int64(8), object(5)
memory usage: 5.3+ MB
```

	0
num_passengers	0
sales_channel	0
trip_type	0
purchase_lead	0
length_of_stay	0
flight_hour	0
flight_day	0
route	0
booking_origin	0
wants_extra_baggage	0
wants_preferred_seat	0
wants_in_flight_meals	0
flight_duration	0
booking_complete	0

dtype: int64

```
# Convert flight_day to numeric
df['flight_day'] = df['flight_day'].map({
    'Mon':1, 'Tue':2, 'Wed':3,
    'Thu':4, 'Fri':5, 'Sat':6, 'Sun':7
})

# Create weekend feature
df['is_weekend'] = df['flight_day'].apply(lambda x: 1 if x>=6 else 0)
```

```
df = pd.get_dummies(df, drop_first=True)
```

```
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier

X = df.drop("booking_complete", axis=1)
y = df["booking_complete"]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

model = RandomForestClassifier(random_state=42)
model.fit(X_train, y_train)
```

```
▾ RandomForestClassifier
RandomForestClassifier(random_state=42)
```

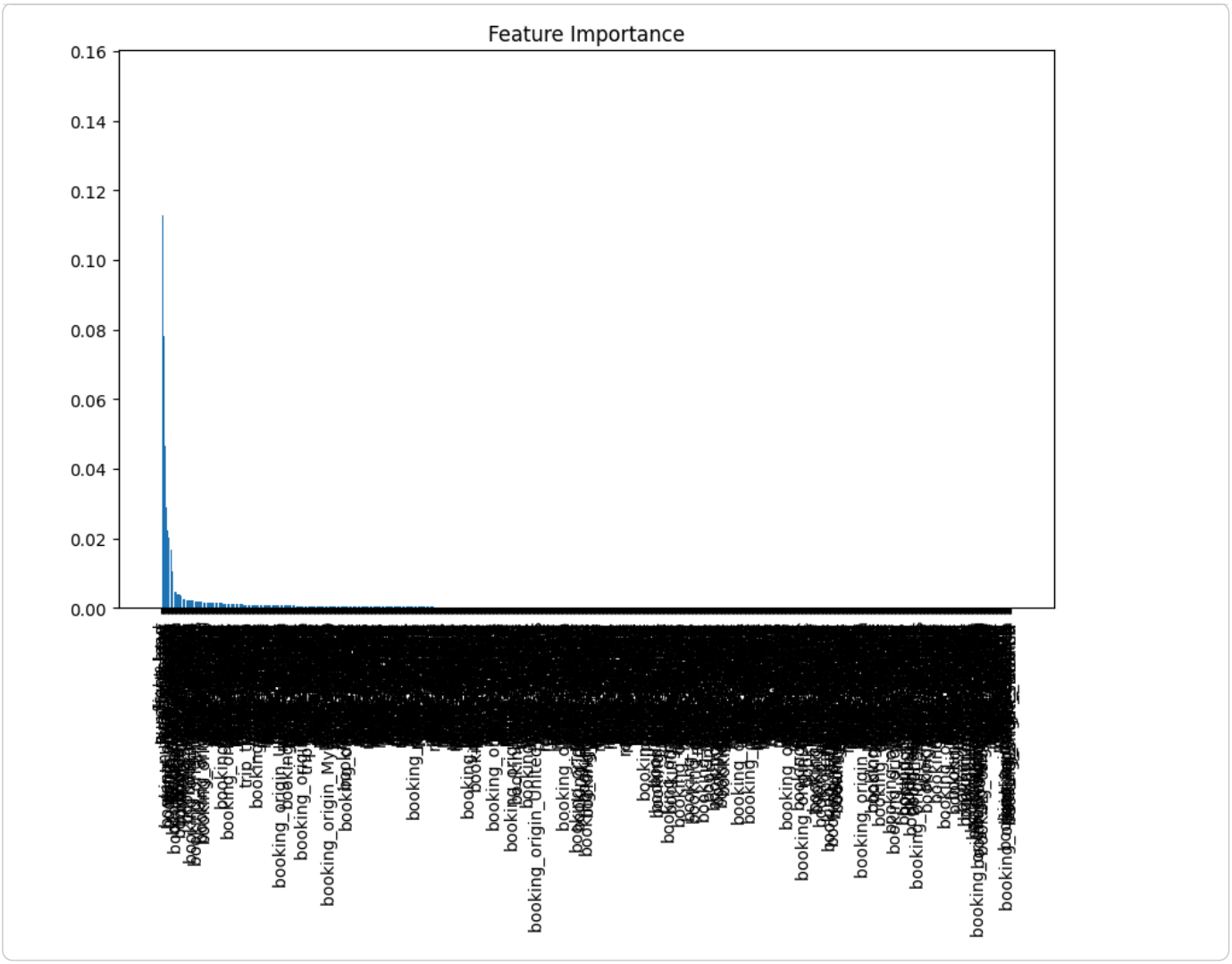
```
from sklearn.model_selection import cross_val_score

scores = cross_val_score(model, X, y, cv=5)
print("Cross Validation Accuracy:", scores.mean())
```

```
Cross Validation Accuracy: 0.71438
```

```
from sklearn.metrics import classification_report, confusion_matrix

y_pred = model.predict(X_test)

print(confusion_matrix(y_test, y_pred))
print(classification_report(y_test, y_pred))
```

```
[[8336  184]
 [1286  194]]
              precision    recall  f1-score   support

           0       0.87      0.98      0.92      8520
           1       0.51      0.13      0.21      1480

    accuracy                           0.85     10000
   macro avg       0.69      0.55      0.56     10000
weighted avg       0.81      0.85      0.81     10000
```

```
import matplotlib.pyplot as plt
import numpy as np

importances = model.feature_importances_
indices = np.argsort(importances)[::-1]

plt.figure(figsize=(10,6))
plt.title("Feature Importance")
plt.bar(range(X.shape[1]), importances[indices])
plt.xticks(range(X.shape[1]), X.columns[indices], rotation=90)
plt.show()
```


Feature Importance
0.16
0.14
0.12
0.10
0.08
0.06
0.04
0.02
0.00